

(In)sensitivity to surface-level heuristics: A case from Turkish verbal attractors

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Abstract

Linguistic illusion literature debates what information accesses memory representations. Prior work tests whether structural, semantic, or discourse cues guide subject-verb dependencies; however, it remains unclear whether native speakers rely on surface level heuristics, such as phonological information during dependency resolution. Traditionally, accidental phonological resemblance to plural ending (e.g., the /s/ in *cruise*) does not induce erroneous agreement in English, whereas resemblance correlating with controllerhood amplifies attraction across various languages. Contradicting this generalization, Slioussar (2018) proposed that accidental phonological resemblance can mediate memory search for Russian subjects. Given the theoretical importance of this proposal and the lack of comparable effects in other languages such as Czech, we propose re-interpret previous findings under the light of a recently growing literature of association with being a possible controller. We test whether phonological overlap or association with controllerhood elicits erroneous agreement in Turkish. Turkish provides a critical test: both verbal and nominal elements can surface as subjects and the plural morpheme *-lar* marks number in both of them, but only nominal plural *-lar* controls verbal agreement. Two speeded acceptability studies show no attraction from plural-marked verbs ($N = 80$; $N = 95$) but robust attraction from genitive plural nouns. We report a first-of-its-kind dissociation under minimal manipulation: verbal attractors that can surface as subjects yet cannot control agreement do not induce attraction, whereas genitive plural nouns—which can be subjects and control in other environments—do. This pattern constrains retrieval processes by tying attraction to abstract controller features rather than surface phonology.

1 Introduction

Human sentence processing draws both on abstract grammatical features and heuristics that exploit surface regularities, such as plausibility (Speer and Clifton, 1998), frequency (Lau et al., 2007), and task-specific factors (Laurinavichyute and von der Malsburg, 2024; Arehalli and Wittenberg, 2021; Hammerly et al., 2019; Logachev and Vasishth, 2016). We focus on one such heuristic: over-reliance on surface form, evidenced when phonological similarity between sentence constituents is observed to modulate performance (Acheson and MacDonald, 2011; Kush et al., 2015; Copeland and Radvansky, 2001; Rastle and Davis, 2008).

A substantial body of work has shown that the parser and the production system are sensitive not only to syntactic or semantic relations but also to the surface form of words. These effects have been taken to

suggest that, under certain circumstances, speakers and comprehenders rely on shallow or heuristic cues to complete dependencies. Acheson and MacDonald (2011), for example, found that participants showed slower reading times when the subjects of the two embedded clauses share phonological similarity (*baker-banker* in 1 vs. *runner-banker* in 2). Moreover, participants were less accurate in answering comprehension questions with phonological overlap present. Related work in short-term memory and word recognition shows similar effects—items that overlap phonologically or morphologically are more confusable and more easily retrieved (Copeland & Radvansky, 2001; Rastle & Davis, 2008).

(1) The baker that the banker sought bought the house.

(2) The baker that the runner sought bought the house.

However, it is unresolved whether this heuristic penetrates dependency resolution itself—including subject-verb agreement, pronoun resolution, or the licensing of negative polarity items—beyond general effects on reading ease and memory. A central question for understanding human cognition is what information is encoded and later available in memory during such dependency resolutions and how faithful these encodings are to the input. Errors in subject-verb agreement have been treated as a key domain for identifying mechanisms of linguistic representation and retrieval (Bock and Miller, 1991; Jäger et al., 2017; Smith and Vasishth, 2020; Phillips et al., 2011). Classic findings demonstrate systematic errors in establishing number agreement between a verb and its agreement controller when an NP with a different number (the attractor) interferes. Speakers produce sentences like (3) or misclassify them as acceptable (Bock and Miller, 1991; Pearlmutter et al., 1999).

(3) * The player on the courts are tired from a long-game.

(4) The players on the courts are tired from a long-game.

Some accounts argue that detailed analyses are not always maintained when heuristics suffice, creating the opportunity for surface regularities to affect judgments (Ferreira et al., 2002; Futrell et al., 2020). For example, lossy-compression-style approaches assume that comprehenders maintain only an imperfect representation of the linguistic input, and that the parser relies on statistical regularities within the language to fill in the gaps

(Futrell et al., 2020). In this model, the input as in (3) would be altered to a more statistically predictable form such as (4), which would be accepted as grammatical. On the other hand, many rational accounts of sentence processing argue that comprehenders maintain detailed and faithful representations of the input (Bock and Miller, 1991; Lewis and Vasishth, 2005). More specifically, cue-based retrieval approaches hold that constituents are stored with detailed abstract features and later accessed by matching retrieval cues, and that erroneous parses can occur when features conflict or interfere. In such accounts, the parser would maintain a detailed representation of the singular subject *player* and the plural attractor *courts*, and the error arises because the retrieval cues for agreement (+PL, +CONTROLLER) match both the singular subject and the plural attractor, leading to interference. However, it remains open whether phonological codes are used as such cues during syntactic dependency building (Lewis and Vasishth, 2005).

Despite much research on what factors modulate agreement errors, the role of phonology remains unclear. The few studies that bear directly on subject-verb agreement exhibit contradictory findings (Bock and Eberhard, 1993; Slioussar, 2018; Lacina and Chromý, 2022). Pseudoplural attractors whose final phone matches the plural suffix and string-ambiguous with other nouns (e.g. *cruise* vs. *crews*) do not increase agreement errors in production (Bock and Eberhard, 1993). Phonological overlap effects have been observed in other cases, but many of them involve additional shared morphological features such as case ambiguity with the controller in the sentence (Hartsuiker et al., 2003; Lago et al., 2019; Lacina and Chromý, 2022), although not all (Slioussar, 2018).

This raises the possibility that surface form affects the formation of agreement dependencies not directly through the use of number form as a retrieval cue, but indirectly, when the surface form is one that is more likely to be realized on agreement controllers.

We test this hypothesis by utilizing the surface-form overlap between the verbal and nominal morphological reflexes of agreement in Turkish. Turkish uses the same surface suffix, *-lar*, for plural marking on nouns and for plural agreement on finite verbs. Crucially, strings bearing verbal *-lar* can occur in subject position, yet they never control finite clause agreement; only nominal plurals do. These properties allow us to test whether form overlap is sufficient to drive attraction, or if the attractor must also be a possible controller

(true of nouns but not verbs). Across two high-powered speeded acceptability experiments in Turkish we find that plural marking on an embedded verbal attractor does not increase acceptance of plural agreement on the matrix verb; such effects are only observed when the plural marker appears on a non-subject noun attractor. These results indicate that surface-form overlap alone does not function as a retrieval cue for agreement in Turkish. Dependency resolution relies on abstract features and structural relations, with phonology influencing processing primarily outside of retrieval.

In the rest of the introduction, we review the role of certain surface cues in agreement attraction, namely case syncretism and pure phonological overlap. We then introduce the previous Turkish attraction studies and brief sketch of the grammar that is relevant for our study. Finally, we present the current study and its predictions.

1.1 Background

It is reported that native speakers from 296 of out 378 languages surveyed exhibit systematic agreement between the verb and another constituent(s), such as subject, object, or both (Siewierska, 2013). However, this agreement process is not always error-free. In their seminal work, Bock and Miller (1991) demonstrated that participants systematically produce erroneous verb forms (*are*) when there is a nearby noun, an attractor, that has a mismatching number as in (5b) compared to their counterpart with singular attractor as in (5a). The effect of the number mismatching attractor, agreement attraction, was also found to be robust in comprehension (Nicol et al., 1997; Pearlmutter et al., 1999) of such sentences in various languages, including Arabic (Tucker et al., 2015), Armenian (Avetisyan et al., 2020), Hindi (Bhatia and Dillon, 2022), Spanish (Lago et al., 2015), Russian (Slioussar, 2018), and Turkish (Lago et al., 2019; Türk and Logačev, 2024; Ulusoy, 2023).

(5) a. Singular Attractor

The player on the court ...

b. Plural Attractor

The player on the courts ...

Many studies have investigated the various syntactic and semantic factors which make agreement errors more likely, which include hierarchical distance (Hartsuiker et al., 2001; Nicol et al., 1997; Kaan, 2002), linear distance (Bock and Cutting (1992); but see Pearlmutter (2000) and Kwon and Strut (2019)), semantic interactions of nouns involved (Eberhard, 1999; Vigliocco et al., 1995; Humphreys and Bock, 2005), and syntactic category of the phrase containing the attractor (Bock and Miller, 1991; Bock and Cutting, 1992). One widely accepted set of accounts that explained these errors are called retrieval based theories (Lewis and Vasishth, 2005; Wagers et al., 2009; Yadav et al., 2023). In these accounts, comprehenders maintain faithful linguistic representations; errors arise because the memory mechanisms used to identify the agreement controller mislead them. Under this approach, phrases are encoded in content-addressable memory as *chunks*—bundles of features including number, gender, and syntactic properties (Smith and Vasishth, 2020). Comprehenders predict the number of the verb based on the noun phrases they process while reading the previous noun phrases. In grammatical sentences with singular verb agreement, the number prediction and the verb number match, which causes no processing difficulty. In contrast, when participants fail to find the predicted number morphology on the verb, a memory-retrieval process is initiated. This process activates the search for a chunk matching relevant cues for agreement controller.

1.1.1 Surface Heuristics: case syncretism and phonological overlap

What are the characteristics of cues which are found useful to be encoded? One line of work manipulated overt case marking on attractors, i.e. syncretism, to test whether morphophonological case is used for dependency resolution. Two grammatical forms are said to be syncretic if they are realized with the same overt morphology despite bearing different syntactic and semantic features. E.g., most noun phrases in English – such as *the cabinet* – are syncretic between nominative and accusative case marking. An exception are some pronouns which differ in their nominative and accusative forms, as with the first person pronoun *I* (nominative) vs. *me* (accusative).

This effect of case syncretism was tested in various languages by manipulating the overt case marking of controllers or attractors, reasoning that surface ambiguity could enhance competition during retrieval or interfere in production. For example, Hartsuiker et al. (2003) in a preamble completion task experiment

used the overlap between accusative and nominative forms of feminine determiners in German and compared these ambiguous forms to distinctively marked dative forms. Participants produced more agreement errors when the preambles contained two noun phrases whose determiners were not distinctively marked, as in (6a), compared to cases where the attractor could be distinguished by form alone, as in (6b). Crucially, this additive effect was limited to feminine nouns, the only gender showing nominative–accusative syncretism in plural forms while other nouns showed the base effect of plural.

- (6) a. Die Stellungnahme gegen die Demonstration-en
the.F.NOM.SG position against the.F.ACC.PL demonstration-PL
‘The position against the demonstrations’
- b. Die Stellungnahme zu den Demonstration-en
the.F.NOM.SG position on the.F.DAT.PL demonstration-PL
‘The position on the demonstrations’

Parallel results were found in comprehension studies in Czech. Chromý et al. (2023) conducted self-paced reading experiments manipulating the number of the attractor and the number of the verb across various syntactic configurations. They found that the attraction effects, i.e. faster reading times at the ungrammatical verb when the attractor is plural, were only observed when the attractor was bearing a case syncretic with the nominative case as in (7), while finding no attraction effects in other experiments where the attractor was unambiguously marked. In a follow up experiment, Lacina et al. (2025) found similar effects in gender agreement within Czech, based on earlier Slovak production findings (Badecker and Kuminiak, 2007). They found a clear gender attraction effect, i.e. faster reading times at the ungrammatical verb when the attractor is of the same gender with the verb, but the head noun was not. More importantly, this effect was only present in cases where the attractor was syncretic in case-marking with the nominative case, which is the case of the agreement controller, but not when the attractor was unambiguously marked with a different case.

- (7) * Složk-a pro archivářk-y nejspís bud-ou zahrnovat veškere
file-NOM.SG for archiver-ACC.PL=NOM.PL probably will-NOM.PL include all
nálezy.
findings
‘A file for archivers will probably include all findings.’

Similarly, Slioussar (2018) found the effects of syncretism in Russian in both production, self-paced readings, and acceptability judgments. She compared sentences with genitive plural attractors, which are unambiguously marked, to sentences with accusative plural attractors, which are syncretic with nominative plural forms, while manipulating other factors such as the number of the attractor, the grammaticality of the sentence, and the presence of a singular subject. She found that sentences with accusative plural attractors yielded more plural completions, faster reading times at the plural verb and higher rates of acceptability compared to the sentences with unambiguous genitive plural attractors.

However, results from other case-marking languages are mixed. For instance, Franck et al. (2010) used French and compared the unambiguously accusative marked attractors to NPs with no overt case marking. They showed that when unambiguous marking increased the attraction effects substantially, contrary to the predictions of cue based retrieval. Avetisyan et al. (2020) observed that unambiguous case in Armenian modulated neither reading times nor error rates. Conversely, Lacina et al. (2025) found that attraction in Czech surfaced only when case morphology was ambiguous. These findings suggest that distinct case morphology is insufficient to predict interference, implicating language-specific distributions or heuristic processing.

The studies discussed above tested the effects of case syncretism, which is a morphophonological overlap that also correlates with the possibility of being a controller. In these cases, while the case of the attractor is ambiguous its number is not. Take the English word *cabinets* as an example. It is syncretic between nominative and accusative case, meaning that its surface form would not change depending on the syntactic case is assigned to it. However, it is not syncretic in number, as the plural form *cabinets* is distinct from the singular form *cabinet*, and this difference would surface as reflex on grammaticality in syntactic and semantic configurations where a certain number is expected, such as **few cabinet* and **a cabinets*.

A second line of work related to surface cues tests a case of accidental phonological overlap that does not itself change the relevant cues. Bock and Eberhard (1993) tested whether attractors that only sound plural, pseudoplural singular attractors such as *cruise* as in (8), increase agreement errors compared to true plural nouns, such as *crews* in (5b). They reasoned that if participants rely on phonological cues rather than

abstract number features, words ending with plural-like sounds (/s/ or /z/) should behave like true plurals. In their preamble completion study, they found that pseudoplural attractors did not induce agreement errors, which argues against a purely phonology-driven account of attraction in English.

(8) The player on the cruise ...

(9) The player on the crews ...

In contrast, [Slioussar \(2018\)](#) reported a contribution of surface-form overlap to agreement in Russian. Recall that she compared genitive and accusative marked attractors. In Russian, a subset of genitive singular nouns (10) is homophonous with nominative plural forms, while genitive plural forms (11) are not ambiguous in this way. In the experiments we previously mentioned, she found that sentences with genitive singular attractors whose form overlaps with nominative plural yielded more plural completions, faster reading times at the plural verb and higher rates of acceptability compared to the sentences with unambiguous genitive plural attractors. She took her results as evidence for a retrieval process in which the search for a controller is mediated through phonological form and relevant features like +NOM and +PL can be activated by simply a phonological overlap.

(10) Komnata dlja večerinki byli ...
 room.NOM.SG for party.GEN.SG=NOM.PL were
 ‘The room for parties/party were ...’

(11) Komnata dlja večerinok byli ...
 room.NOM.SG for party.GEN.PL≠NOM.PL were
 ‘The room for parties/party were ...’

However, another Slavic language Czech which shows the same ambiguity between the genitive singular and nominative plural forms was found to not show attraction effects by simple phonological overlap ([Lacina and Chromý, 2022](#)). These mixed findings in case-syncretism literature, English pseudoplural, and a failure to replicate in another Slavic language cast a shadow on phonological modulation explanation.

1.1.2 Accounting for syncretism and phonological-overlap

To the best of our knowledge, there is no attempt of explanation within Rational Accounts or Marking and Morphing accounts to explain the variety of findings in the literature regarding surface cues in agreement attraction. Below we outline existing models of the syncretism effects under a cue-based approach, and then how other account would approach these findings.

In canonical cue-based retrieval accounts, a dependency is formed via a content-addressable search for a relevant *chunk*, i.e. bundles of features for a given lexical item, that matches the retrieval cues provided by the verb (Lewis and Vasishth, 2005). When a verb encountered, a search is triggered for a chunk that matches the necessary features for agreement. In the case of English subject-verb agreement, the verb would provide cues such as +NOM and +PL, and the search would activate chunks that match these cues. If there is a singular subject and a plural attractor, both of which match the +NOM cue but only the plural attractor matches the +PL cue, then the retrieval process can be misled by the attractor, leading to agreement errors.¹

In this account, the case syncretism is expected to lead to difficulty in correctly determining the head due to the increased competition between chunks that shares relevant cues. On the other hand, when the attractor is not case-syncretic, the partial match between the retrieval cues and the attractor would be weaker, leading to less attraction effects (Smith and Vasishth, 2020; Yadav et al., 2023).

Slioussar (2018) situated her findings within the cue-based retrieval accounts. She argued that the phonological overlap between the genitive singular and nominative plural forms can activate relevant features like +NOM and +PL in addition to +GEN and +SG, which would increase the likelihood of retrieving the attractor as a controller, leading to more agreement errors in cases where the attractor only matches with the +GEN and +PL cues (10). To the best of our knowledge, this is the only account that directly exploits the co-activation of chunks through phonological overlap to explain the effects of syncretism. However, Slioussar’s (2018) mechanism aligns closely with foundational principles of word recognition and working memory. Models of continuous speech parsing (e.g., McClelland and Elman, 1986, Norris (1994)) have long established that phonological string overlap automatically triggers the transient activation of embedded or

¹Lewis and Vasishth (2005) argue that the retrieval cues for case is not determined by the syntactic position or the abstract case features, but rather by the surface form of the noun phrase.

competing lexical chunks (Shillcock, 1990). Furthermore, the broader working memory literature demonstrates that phonological similarity creates significant interference during retrieval (Baddeley, 1966; Conrad, 1964; Acheson and MacDonald, 2009)

However, the same surface-form overlap did not give rise to attraction effects in Czech, another Slavic language (Lacina and Chromý (2022)). These mixed findings in case-syncretism literature, English pseudoplural, and a failure to replicate in another Slavic language cast a shadow on phonological modulation explanation.

An alternative account posits that attraction errors arise not from phonological co-activation of competing parses, but from the use of language-general statistical heuristics. Under this view, comprehenders probabilistically associate certain surface cues—word order, case syncretism, or the presence of certain morphemes—with controllerhood, the property of being a possible agreement controller. In cases of syncretism, then, certain noun phrases might carry an increased association with controllerhood due to the distribution of such forms in the language. For example, Lago et al. (2019) argue that Turkish speakers retrieve genitive-marked attractors as controllers because genitive case controls agreement in embedded clauses, even though it cannot do so in matrix clauses. The syncretism between the nominal modifier and the embedded subject is thus phonological rather than functional, and attraction arises because Turkish speakers associate genitive-marked nominals with being controllers.

Converging evidence for this sensitivity to “looking like a controller” comes from Romanian and Hindi (Bhatia and Dillon, 2022; Bleotu and Dillon, 2024). Bleotu and Dillon (2024) found that Romanian attractors induced agreement errors only when they surfaced with a determiner, as opposed to bare forms. Since only nouns bearing a determiner can control agreement in Romanian, they argue that participants associate the presence of a determiner with controllerhood. In Hindi, Bhatia and Dillon (2022) found that plural-marked attractors were erroneously retrieved as controllers only when the attractor also served as an agreement controller within the embedded clause—independent of whether its syntactic role was that of an object or a subject. They argue that participants track controllerhood within a sentence rather than relying on language-general distributional statistics alone. Nevertheless, this finding also demonstrates that agreement processes are sensitive to the abstract feature of being a controller.

Further evidence comes from English. In a series of six experiments, [Schlueter et al. \(2018\)](#) showed that the coordinator *and* when coordinating two singular noun phrases induces attraction even in the absence of overt plural morphology *-s*, because they argue *and* is statistically associated with plurality. Crucially, not only conjoined singular noun phrases lacking overt plural morphology are good candidates for attraction, but the coordinator *and* alone can induce the effect when it conjoins two adjectives modifying a singular noun (e.g., *the slogan about the loyal and caring husband*). They argue that participants exploit the statistical association between *and* and plurality, which leads them to accept ungrammatical sentences containing a plural auxiliary. Taken together, these explanations across languages and structures suggest that the cue-chunk match is not strictly categorical but can be influenced by statistical associations within a language ([Engelmann et al., 2019](#)). Importantly, the Hindi findings indicate that the human parser is sensitive to the abstract notion of being a controller, and not merely to language-general co-occurrence statistics.

A similar account extends to Russian. While genitive-marked nouns can serve as subjects in negative inversion constructions, they do not control verbal agreement in these contexts. Crucially, however, they remain active controllers within the noun phrase, triggering number or gender marking on modifiers (e.g., surfacing as feminine *ni odnoy* with a feminine head, contrasting with masculine *ni odnogo* in 12) ([Babby, 2001](#); [Partee and Borschev, 2004](#)). In contrast, Czech does not allow genitive subjects, and thus not license these controller properties in subject positions.

(12) ..., tam ne rabotaet ni odnogo inženera.
 ..., there NEG works not one.M.SG.GEN engineer.M.SG.GEN
 ‘..., there hasn’t been a single engineer working there.’

1.1.3 Sketch of Turkish and Attraction in Turkish

Turkish offers a useful test case because genitive-marked nominals can carry controller-like cues in some structures, while verbal agreement is morphologically rich and overt. In genitive-possessive NPs (roughly analogous to English Saxon genitives), the possessor bears genitive case and the head noun bears possessive morphology.

Using this construction, [Lago et al. \(2019\)](#) reported robust attraction: participants accepted ungrammatical

plural agreement more often when the possessor was plural (13) than when it was singular (14).

- (13) Teknisyen-ler-in eğitmen-i olağanüstü hızlı koş-tu-lar.
technician-PL-GEN instructor-POSS extraordinary fast run-PST-PL

‘The technicians’ instructor ran_{PL} extraordinarily fast.’

- (14) Teknisyen-in eğitmen-i olağanüstü hızlı koş-tu-lar.
technician-GEN instructor-POSS extraordinary fast run-PST-PL

‘The technician’s instructor ran_{PL} extraordinarily fast.’

Türk and Logačev (2024) asked whether this attraction might be partly driven by a local parsing ambiguity in the original materials. In Lago et al. (2019), many head nouns were consonant-final, so the head suffix *-i* was syncretic between 3sg possessive and accusative. This creates a potential misparse in which the genitive possessor is temporarily treated as a clause-level subject and the head as an object, which could artificially increase the possessor’s controller-like status.

To test this ambiguity account, Türk and Logačev (2024) used vowel-final heads that disambiguate the two morphemes: 3sg possessive surfaces as *-si*, whereas accusative surfaces as *-yi*. The logic was straightforward: if attraction in Turkish genitive-possessive NPs mainly comes from this local ambiguity, then attraction should be substantially reduced when the head morphology is unambiguous.

The attraction effect persisted. Plural genitive possessors still increased ungrammatical acceptability, and the magnitude was comparable to Lago et al. (2019). This result argues against an ambiguity-only explanation and suggests that genitive-marked possessors carry controller-relevant cues independently of that local form overlap.

Ulusoy (2023) extended this literature to configurations where the attractor is not in the same phrase as the controller. Using matrix-clause subjects as attractors for embedded verbal agreement (15), she found more errors with plural than singular attractors. At the same time, ungrammatical acceptance was relatively high in both conditions, despite cross-linguistic evidence that such structural separation usually reduces attraction (Bock and Cutting, 1992; Franck et al., 2002). Taken together, Turkish studies establish reliable attraction with plural genitive cues, but they leave open which cues are doing the work: surface form, controllerhood, or both.

308 (15) * Kütüphaneci-ler [çalışan öğrenci-nin iste-dik-ler-i] kitab-ı şimdi
 309 librarian-PL hardworking student.SG-GEN want-NMLZ-3SG-POSS book-ACC now
 find-du-lar.
 find-PST-3PL
 310 ‘The librarians found the book that the hardworking student wanted now.’

311 1.2 This study

312 Motivated by these alternative accounts and conflicting findings (Bock and Eberhard, 1993; Lacina and
 313 Chromý, 2022) along with the theoretical importance of such proposal, we test the phonological modulation
 314 hypothesis in two high-powered experiments: whether a syntactically ineligible controller, but still a possible
 315 subject, can induce attraction solely through morphophonological overlap matching the agreement suffix in
 316 form and semantics. To this end we capitalize on the shared surface form of verbal and nominal plural
 317 marking (-*lar*) in Turkish to target this question. We use reduced relative clauses (RRCs) where the plural-
 318 marked verb appears as the attractor (16). Crucially, this -*lar* syncretism is not feature-ambiguous; it is a
 319 form-only overlap lacking the possibility of being a potential controller. Even when a headless RRC alone
 320 surfaces as a subject, it cannot control agreement (17).

321 (16) Gör-dük-ler-i çocuk koş-tu-(*lar).
 go-NMLZ-PL-POSS kid[NOM] run-PST-(*PL)
 322 ‘The kid that (they) saw ran.’

323 (17) Gör-dük-ler-i koş-tu-(*lar).
 go-NMLZ-PL-POSS run-PST-(*PL)
 324 ‘(The kid) that (they) saw ran.’

325 In Experiment 1, we tested the form hypothesis by comparing sentences with verbal attractors to sentences
 326 with canonical nominal attractors in Turkish. Experiment 2 then tested the form hypothesis more directly by
 327 only using verbal attractors. We expected that if surface-overlap can modulate relevant memory representa-
 328 tions for dependency resolutions, we would see similar attraction results with nominal and verbal attractors.
 329 However, if participants are tracking an higher order cue that is relevant for being a possible controller, then
 330 the verbal attractors, due to their inability to control agreement, would not introduce agreement attraction
 331 effects even though their high morpho-phonological similarity.

Across both experiments, we found no evidence that verbal *-lar* induces attraction, even when canonical nominal attractors are present in the same session. This pattern aligns with prior findings in general attraction literature and Turkish agreement attraction, namely surface-form overlap alone does not derive agreement illusions. Rather, attraction appears to depend on abstract feature overlap between potential controllers and agreement probes, and possibly statistical associations between the strings and their controllers. In this light, findings of Slioussar (2018) are best analyzed as a possible increased association between genitive marking and possible subjecthood and being an agreement controller, which is not possible in Czech, and thus no attraction (Lacina and Chromý, 2022). By doing so, we hope to clarify how cue-mechanisms are employed and the role of phonological overlap in sentence processing.

2 Experiment 1: Testing Surface-Form Overlap

2.1 Participants

We recruited 95 undergraduate students to participate in the experiment in exchange for course credit. All participants self-identified as native Turkish speakers, with an average age of 21 years (range: 18-30).

Subject-level screening used two discrimination checks. We excluded participants whose accuracy was below 25% in either Nominal or Relative Clause conditions with singular attractor and singular-marked verb. 1 participant were below threshold in Nominal conditions, and 2 participant(s) were below threshold in Relative Clause conditions, resulting in 3 unique participant(s) excluded at subject level (3.16% of the recruited sample). This subject-level filtering removed 240 rows before trial-level trimming.

At the trial level, response-time trimming removed 171 trials (111 with $RT \leq 200$ ms; 60 with $RT \geq 4999$ ms; 2.32% of trials). After RT trimming, there were no missing responses. In total, 5.41% of collected trials were excluded, and the final dataset contained 92 participants and 7189 observations.

2.2 Materials

We used 40 sets of sentences like Table 1, in which we manipulated (i) the number of the attractor, (ii) the type of the attractor, and (iii) the number agreement on the verb. Both plural markings were marked with

the suffix *-lar*, while the singular number and singular agreement were marked by its absence.

Table 1: Experimental conditions. The Attractor was manipulated for number and type. The Verb was manipulated to match or mismatch the head noun (always singular), creating Grammatical and Ungrammatical conditions.

Attr. Type	Attr. Num	Attractor	Grammaticality (Verb Suffix)	
			Grammatical	Ungrammatical (*)
Verbal	SG	Tut-tuğ-u <i>hire-NMLZ-POSS</i>	zıpla-dı <i>jump-PST</i>	*zıpla-dı-lar <i>jump-PST-PL</i>
	PL	Tut-tuk-lar-ı <i>hire-NMLZ-PL-POSS</i>	zıpla-dı <i>jump-PST</i>	*zıpla-dı-lar <i>jump-PST-PL</i>
Nominal	SG	Milyoner-in <i>millionaire-GEN</i>	zıpla-dı <i>jump-PST</i>	*zıpla-dı-lar <i>jump-PST-PL</i>
	PL	Milyoner-ler-in <i>millionaire-PL-GEN</i>	zıpla-dı <i>jump-PST</i>	*zıpla-dı-lar <i>jump-PST-PL</i>

(18) *Verbal Attractor Conditions*

[**Attractor**] aşçı mutfak-ta sürekli [**Verb**]
 hire-NMLZ-(PL)-POSS cook kitchen-LOC non.stop jump-PST-(PL)
 ‘The [_{Attr.} **hired_{pl}**/**hired_{sg}**] cook [_{Verb} **jumped_{pl}**/**jumped_{sg}**] in the kitchen non-stop.’

(19) *Nominal Attractor Conditions*

[**Attractor**] aşçı-sı mutfak-ta sürekli [**Verb**]
 millionaire-(PL)-GEN cook-POSS kitchen-LOC non.stop jump-PST-(PL)
 ‘The [_{Attr.} **millionaires**’/**millionaire’s**] cook [_{Verb} **jumped_{pl}**/**jumped_{sg}**] in the kitchen non-stop.’

Verbal attractor conditions featured complex subject NPs containing a bare head noun and a reduced relative clause acting as the attractor (e.g., ‘tuttukları aşçı’, ‘the hired cook’). Because nominal plural marking is mandatory and the head noun was always singular, plural verb agreement rendered these sentences ungrammatical. Nominal attractor conditions, featuring nominal attractors such as ‘milyonerlerin aşçısı’ (‘the millionaires’ cook’) were taken from [Türk and Logačev \(2024\)](#). To prevent participants from associating plural verbs with ungrammaticality, fillers were balanced between grammatical sentences with plural verbs and ungrammatical sentences with singular verbs.

2.3 Procedures

The experiment was conducted online via Ibex Farm ([Drummond, 2013](#)), lasting approximately 25 minutes. After providing informed consent and demographic details, participants read instructions and completed nine practice trials.

Each trial began with a 600 ms blank screen, followed by a centered, word-by-word RSVP presentation (30 pt font, 400 ms duration, 100 ms inter-stimulus interval). Upon the prompt, participants judged sentence acceptability as quickly as possible by pressing ‘P’ (acceptable) or ‘Q’ (unacceptable). A red warning message appeared during practice trials—but not experimental trials—if responses exceeded 5,000 ms. Participants pressed the space bar to advance to the next item.

The study included 40 experimental and 40 filler sentences. Experimental items were distributed across four lists using a Latin-square design, ensuring each participant viewed only one list containing one version of each item.

2.4 Data Availability

Materials, code and data for both experiments available at: https://osf.io/p6243/overview?view_only=a4b4f0e64d924d48b68782169d646381

2.5 Analysis and Results

Participants showed high accuracy in both grammatical ($M = 0.95$, $CI = [0.94, 0.96]$) and ungrammatical filler sentences ($M = 0.06$, $CI = [0.05, 0.07]$), indicating that they understood the task and performed it reliably.

Figure 1 shows mean ‘yes’ responses with 95% confidence intervals across conditions, together with the nominal-attractor benchmark from Türk and Logačev (2024) (similar in magnitude to Lago et al. (2019)). In the nominal-attractor conditions, we see the expected attraction profile: ungrammatical sentences with plural genitive attractors received more ‘yes’ responses than ungrammatical sentences with singular genitive attractors ($M = 0.11$, $CI = [0.09, 0.15]$ vs. $M = 0.04$, $CI = [0.03, 0.07]$).

This increase does not carry over to verbal relative-clause attractors. Ungrammatical RC items remain near floor in both number conditions (singular: $M = 0.04$, $CI = [0.02, 0.06]$; plural: $M = 0.04$, $CI = [0.03, 0.06]$). Grammatical sentences were generally rated highly across attractor type and number.

Our model-based analysis targeted the same question as the descriptive results: whether verbal attractors

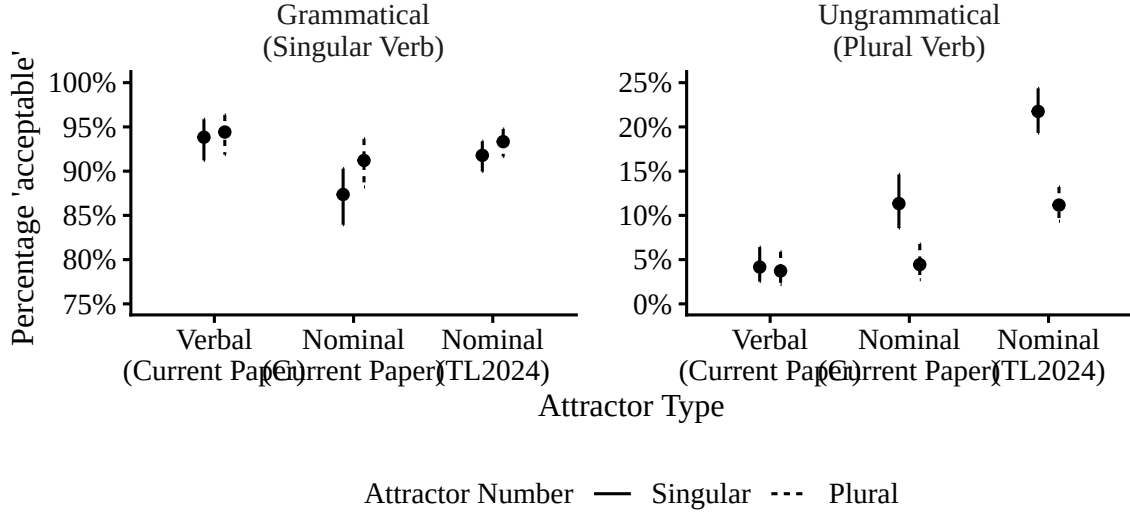


Figure 1: Mean proportion of ‘acceptable’ responses by grammaticality, attractor number and attractor type. Error bars show 95% Clopper–Pearson confidence intervals.

induce attraction. We fitted a Bayesian mixed-effects logistic regression to binary *yes/no* responses, combining the present dataset with the nominal-attractor dataset from Türk and Logačev (2024). We used uninformative (weakly informative) priors for fixed effects ($\mathcal{N}(0, 1)$) and standard regularizing priors for remaining parameters. The $\mathcal{N}(0.85, 0.70)$ prior is on the intercept parameter (log-odds scale), not directly on the observed data. Its central 95% prior interval is approximately $[-0.52, 2.22]$, corresponding to a baseline *yes* probability of about $[0.37, 0.90]$. Full model specifications and priors are reported in Table 2, and the exact contrast coding is reported in Table 3.

Table 2: Bayesian model specifications for Experiment 1.

Specification	Value	Sampling detail	Value
Family (link)	Bernoulli (logit)	Chains	4
Intercept prior	$\mathcal{N}(0.85, 0.70)$	Iterations	12000 (2000 warmup)
Fixed-effect priors	$\mathcal{N}(0, 1)$	Threads	4
Random-effect SD prior	Exponential(1)	<code>sample_prior</code>	“yes”
Random-effect correlation prior	LKJ(2)	Init / seed	0 / 1

Table 3: Contrast coding used in the Experiment 1 Bayesian model (T1 = RC vs average of nominal types; T2 = Gen-Current vs Gen-TL24).

		Grammaticality	Attractor	Type Contrast 1	Type Contrast 2
Factor	Level	(G)	Number (A)	(T1)	(T2)
Grammaticality	Grammatical	0.5	NA	NA	NA
Grammaticality	Ungrammatical	-0.5	NA	NA	NA
Attractor	Plural	NA	0.5	NA	NA
Number					
Attractor	Singular	NA	-0.5	NA	NA
Number					
Attractor	RC	NA	NA	0.333	0.000
Type					
Attractor	Gen-Current	NA	NA	-0.167	0.333
Type					
Attractor	Gen-TL24	NA	NA	-0.167	-0.333
Type					

Table 4 reports posterior summaries for key fixed effects in the full model. The model shows a strong grammaticality effect (G: 6.25 [5.81, 6.71], $P(\text{Beta} > 0) = >0.99$). The critical attraction term is also non-zero overall (G x A: -0.97 [-1.46, -0.49], $P(\text{Beta} > 0) = <0.01$). Most importantly for the hypothesis, the three-way interaction with the verbal-vs-nominal contrast is positive (G x A x T1: 2.41 [0.16, 4.65], $P(\text{Beta} > 0) = 0.98$), indicating that attraction is substantially weaker in verbal RC conditions than in nominal conditions. In contrast, the interaction separating the two nominal datasets is uncertain (G x A x T2: -0.63 [-2.11, 0.85], $P(\text{Beta} > 0) = 0.20$), providing little evidence for a nominal-dataset difference in attraction.

To directly test whether attraction is present in verbal conditions, we fit a model restricted to RC trials. In this follow-up model, the Grammaticality - Attractor Number interaction was -0.15 [-1.04, 0.75] ($P(\text{Beta} < 0) = 0.64$). Because the posterior interval spans zero and the directional probability is not extreme, we

conclude that we do not have any evidence for attraction effects within verbal conditions in Experiment 1.

Table 4: Posterior summaries of key fixed effects in the full Bayesian model for Experiment 1.

Effect	Beta [95% CrI]	P(Beta > 0)
Grammaticality (G)	6.25 [5.81, 6.71]	>0.99
Attractor Number (A)	0.21 [-0.04, 0.45]	0.95
Attractor Type: RC_vs_Gens (T1)	-0.67 [-1.48, 0.15]	0.05
Attractor Type: GenCurrent_vs_GenTL24 (T2)	-1.10 [-1.77, -0.43]	<0.01
G x A	-0.97 [-1.46, -0.49]	<0.01
G x A x T1	2.41 [0.16, 4.65]	0.98
G x A x T2	-0.63 [-2.11, 0.85]	0.20

To quantify evidence for the null effect in the verbal conditions, we compared two RC-only Bayesian mixed-effects models using Bayes Factors estimated by bridge sampling via **brms** (Bürkner, 2017). The main (alternative H) model included the critical Grammaticality - Attractor Number interaction, whereas the null model (H0) omitted this interaction while keeping the same main effects and random-effects structure. Thus, the comparison directly tests whether verbal attractors induce agreement attraction in RC conditions. We report $BF_{01} = p(\text{data} \mid M_0) / p(\text{data} \mid M_1)$, where M_0 is the simpler RC-only model without the interaction and M_1 is the RC-only model with the interaction. Under equal prior odds for the two models, BF_{01} expresses how much more likely the observed data are under the null model than under the interaction model. In the present data, $BF_{01} = 6.13$, indicating moderate evidence for the null model. Following Jeffreys (1961, as cited in Lee and Wagenmakers (2014)), BF values between 3 and 10 are taken as moderate evidence, values between 10 and 30 as strong evidence, values between 30 and 100 as very strong evidence, and values above 100 as extreme evidence.

Because Bayes Factors depend on the prior distribution assigned to the parameter of interest, we also conducted a prior-sensitivity analysis for the RC-only comparison. Specifically, we refit the same pair of nested models under a range of increasingly concentrated zero-centered normal priors on the fixed effects

($\mathcal{N}(0, 2.0)$, $\mathcal{N}(0, 1.0)$, $\mathcal{N}(0, 0.5)$, $\mathcal{N}(0, 0.25)$, and $\mathcal{N}(0, 0.1)$) and recomputed BF_{01} for each comparison. This analysis asks whether the conclusion that there is no verbal attraction depends on assuming a broader prior that allows large effects, or whether it remains stable as the prior increasingly concentrates mass near zero. We therefore interpret the Bayes-factor results together with this sensitivity table, rather than treating a single prior specification as decisive.

Table 5: Prior-sensitivity analysis for the RC-only Bayes factor comparison. $\text{BF}_{01} > 1$ favors the simpler RC-only model without the Grammaticality - Attractor Number interaction.

Prior Set	Prior Details	Prior Density		BF01 (G	
		at 0	Prior P(beta < 0.20)	x A)	Interpretation
Wide	$\mathcal{N}(0, 2.00)$	0.20	0.08	12.40	strong evidence for null
Main	$\mathcal{N}(0, 1.00)$	0.40	0.16	4.73	moderate evidence for null
Narrow	$\mathcal{N}(0, 0.50)$	0.80	0.31	4.60	moderate evidence for null
Very Narrow	$\mathcal{N}(0, 0.25)$	1.60	0.58	2.73	anecdotal evidence for null
Ultra Narrow	$\mathcal{N}(0, 0.10)$	3.99	0.95	3.00	anecdotal evidence for null

Table 5 can be read from left to right. The first two numeric columns (Prior Density at 0 and Prior P(|beta| < 0.20)) indicate how strongly each prior concentrates mass near no effect, with larger values reflecting a more skeptical prior. The BF01 (G x A) column reports the evidence for the simpler RC-only model without the Grammaticality - Attractor Number interaction, so values greater than 1 favor the null model and values less than 1 favor the interaction model. The Interpretation column maps these values onto the conventional evidence categories following Jeffreys (1961, as cited in Lee and Wagenmakers (2014)). In Table 5, BF_{01} favors the null model for all five priors, ranging from 2.73-12.40; the weakest support is

under the Very Narrow prior and the strongest support is under the Wide prior. This means that the null conclusion is stable in direction across priors, although the strength of evidence varies from anecdotal to strong depending on prior width.

To make this pattern directly interpretable at the condition level, we next derived model-implied attraction and acceptability contrasts from posterior draws of the full Experiment 1 model. Specifically, we extracted the posterior draws for the baseline Grammaticality - Attractor Number interaction ($G \times A$) and the two three-way interaction terms ($G \times A \times T1$, $G \times A \times T2$), and then combined them using the Helmert-coding weights in Table 3 to recover the implied attraction effect separately for RC, Gen-Current, and Gen-TL24 conditions. We summarize each derived contrast by its posterior mean, 95% credible interval, and posterior probability of being negative, where more negative values indicate stronger attraction. These derived contrasts again show strong negative attraction effects for the nominal conditions ($M = -1.59$, $CI = [-2.28, -0.89]$, $P(<0) = >0.99$; $M = -1.16$, $CI = [-1.64, -0.69]$, $P(<0) = >0.99$), little to no attraction for the verbal RC condition ($M = -0.17$, $CI = [-1.01, 0.68]$, $P(<0) = 0.63$), and no clear Gen-Current vs Gen-TL24 difference in attraction ($M = -0.42$, $CI = [-1.25, 0.41]$, $P(<0) = 0.8$). Separately, to assess the shift in baseline acceptability across the two nominal datasets, we extracted the posterior for the main effect of the `GenCurrent_vs_GenTL24` attractor-type contrast from the same full model; this showed lower overall acceptability in the current nominal dataset relative to [Türk and Logačev \(2024\)](#) ($M = -1.1$, $CI = [-1.77, -0.43]$, $P(<0) = >0.99$), indicating a distributional shift in baseline acceptability without a corresponding shift in attraction magnitude.

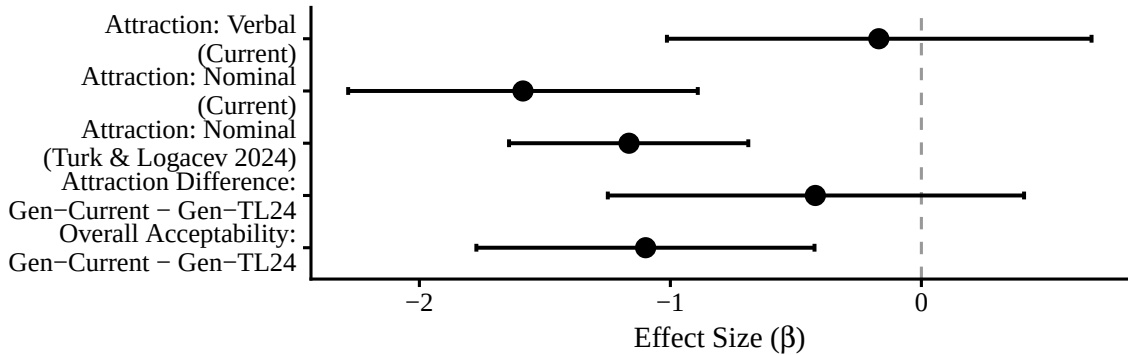


Figure 2: Posterior summaries of attraction-related effects. Points indicate posterior means, and horizontal bars show 95% credible intervals on the log-odds (β) scale. Attraction was estimated as the interaction between grammaticality and attractor number within each attractor type. Negative values indicate stronger attraction (a reduced ungrammaticality penalty in plural-attractor conditions). Dashed line denotes zero (no effect).

2.6 Discussion

Experiment 1 tested whether phonological overlap between nominal and verbal plural marking in Turkish is sufficient to induce agreement attraction. Our results did not show evidence for verbal attraction. Participants reliably rejected ungrammatical sentences with plural-marked verbal attractors, while showing canonical attraction in the nominal-attractor conditions that replicate prior work (Türk and Logačev, 2024; Lago et al., 2019). We confirmed that this dissociation is not an artifact of the mixed design: a follow-up model restricted to verbal-attractor trials showed no interaction between Grammaticality and Attractor Number. Our Bayes Factor and prior sensitivity analysis also suggested that the models without the interactions were preferred over the models with interaction included. This pattern is more consistent with accounts in which attraction depends on controller-relevant cues rather than surface-form overlap alone (Lago et al., 2019); we return to this in the General Discussion.

A potential concern is that our mixed design that combines canonical nominal attractors with verbal ones. The presence of robust nominal-attractor items may have influenced response patterns in ways that obscure genuine verbal attraction. The presence of robust nominal-attractor items could have shifted participant strategies, either masking weaker verbal effects or encouraging direct comparison between the two attractor types. Even though we did not see the reverse of such effect on the attraction in nominal conditions, overall acceptability for nominal items was lower in the present study than in Türk and Logačev (2024), suggesting that the experimental context might have modulated response patterns (we discuss the implications of this distributional shift further in the General Discussion). To determine whether the absence of verbal attraction is genuine or a distributional artifact of the mixed design, Experiment 2 removed all nominal attractors, testing whether the null effect persists when verbal morphology is the sole potential source of interference.

3 Experiment 2: Isolating Verbal Attractors

Experiment 1 showed no attraction from verbal attractors, but the presence of robust nominal-attractor items may have influenced response patterns. Experiment 2 addressed this by removing all nominal-attractor

conditions, testing whether the null verbal effect persists when verbal morphology is the sole potential source of interference.

3.1 Participants

We recruited 80 undergraduate students to participate in the experiment in exchange for course credit. All participants self-identified as native Turkish speakers, with an average age of 21 years (range: 18-31).

Subject-level screening used one discrimination check. Because Experiment 2 contained only verbal attractor conditions, we computed each participant’s difference in acceptance rates between grammatical (singular-verb) and ungrammatical (plural-verb) sentences with singular attractors. We excluded participants whose discrimination score fell at or below 25%. This removed 2 participant(s) (2.5% of the recruited sample).

At the trial level, response-time trimming removed 185 trials (154 with $RT \leq 200$ ms; 31 with $RT \geq 4999$ ms; 2.96% of trials). After RT trimming, there were no missing responses. In total, 5.39% of collected trials were excluded, and the final dataset contained 78 participants and 6055 observations.

3.2 Materials

We used the same verbal attractor items and fillers as in Experiment 1, removing all nominal attractor trials.

3.3 Procedures

The procedure was identical to Experiment 1.

3.4 Analysis and Results

Participants showed high accuracy in both grammatical ($M = 0.94$, $CI = [0.92, 0.95]$) and ungrammatical filler sentences ($M = 0.92$, $CI = [0.9, 0.93]$), indicating that they understood the task and performed it reliably.

Figure 3 presents the overall means and credible intervals for ‘yes’ (acceptable) responses across experimental conditions. As shown, ungrammatical sentences with plural attractors were rated as acceptable as their counterparts with singular attractors ($M = 0.06$ and 0.05 , $CI = [0.04, 0.07]$ and $[0.03, 0.07]$ for singular and

plural attractors, respectively).

On the other hand, accuracy in grammatical conditions was modulated by the number of the attractor in an unexpected way. Participants rated grammatical sentences with singular attractors as grammatical less often ($M = 0.92$, $CI = [0.9, 0.94]$) compared to their counterparts with plural attractors ($M = 0.95$, $CI = [0.93, 0.96]$).

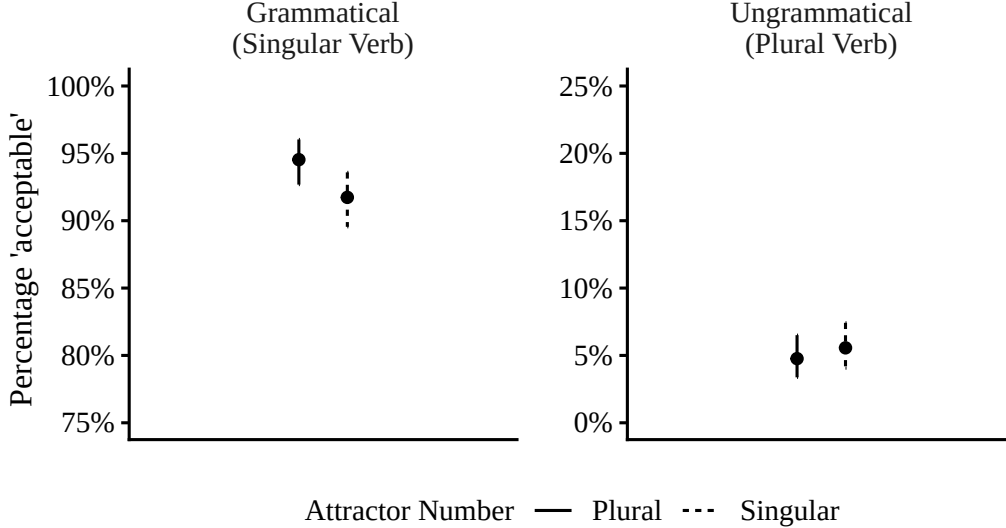


Figure 3: Mean proportion of ‘acceptable’ responses by grammaticality and attractor number. Error bars show 95% Clopper–Pearson confidence intervals.

We fitted a Bayesian mixed-effects logistic regression to binary *yes/no* responses using *brms*, with the same model family (Bernoulli, logit link) and uninformative priors as in Experiment 1. Because Experiment 2 included only verbal attractor conditions, the model had a simpler 2×2 structure: fixed effects for Grammaticality and Attractor Number (both sum-coded at ± 0.5) and their interaction, with maximal random intercepts and slopes for both subjects and items.

Posterior estimates are summarized in Figure 4. The model revealed a positive effect of grammaticality ($\beta = 6.32$ [5.74, 6.97], $P(\beta > 0.99)$), but no reliable main effect of attractor number ($\beta = 0.17$ [-0.22, 0.57], $P(\beta > 0.80)$). The interaction was small but positive ($\beta = 0.86$ [0.06, 1.75], $P(\beta > 0.98)$); crucially, this interaction reflected a grammatical-condition asymmetry rather than attraction. To clarify the pattern within each grammaticality level, we computed model-implied attractor-number effects from posterior draws. In grammatical conditions, plural attractors increased acceptance relative to singular attractors ($\beta = 0.60$

[0.05, 1.18], $P(\beta > 0) = 0.98$). In ungrammatical conditions—the attraction-relevant comparison—there was no evidence for an attractor-number effect ($\beta = -0.27$ [-0.87, 0.31], $P(\beta > 0) = 0.19$). Thus, the presence of a plural verbal attractor did not increase the acceptability of ungrammatical sentences; the positive interaction is driven entirely by a grammatical-condition pattern that is opposite in direction to what an attraction account would predict.

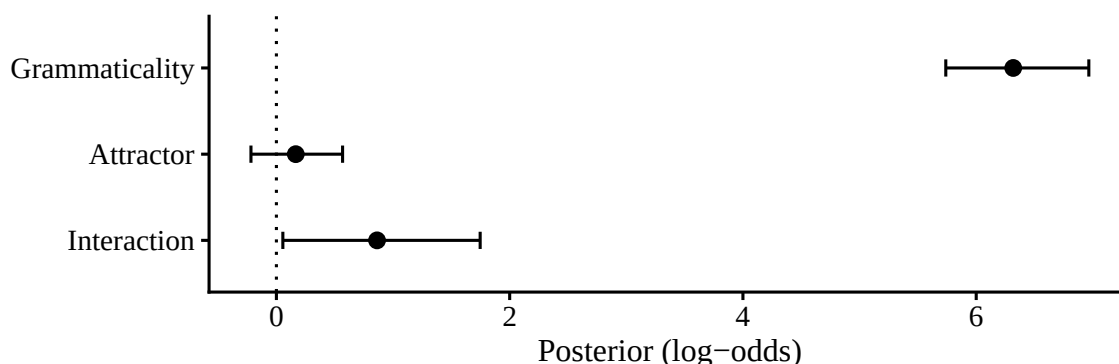


Figure 4: Posterior means and 95% credible intervals for fixed effects in the Experiment 2 Bayesian model (log-odds scale).

3.5 Discussion

Experiment 2 replicated the verbal attractor conditions from Experiment 1 in isolation and again revealed no evidence for agreement attraction driven by plural markers on the verbal attractor. Ungrammatical sentences with plural-marked reduced clause verbs were rejected at similar rates regardless of attractor number. This confirms that the absence of verbal attraction in Experiment 1 was not due to the presence of nominal attractor items or to the distributional properties of the mixed design.

Unexpectedly, grammatical sentences with singular attractors were judged less acceptable than those with plural attractors. This effect is unlikely to reflect agreement attraction, since it arises in the opposite direction from what attraction would predict, meaning singular attractor results in increased rejection of singular verb as grammatical. One possibility is that it results from an interaction between plausibility and referential availability: the plural morpheme can license a more general interpretation, whereas the singular reduced relative clause more strongly invites a specific referent that may be less accessible in the task context. On the other hand, reduced relative clauses without a *had* may act like a correlative structures in Turkish. In

the case of *Gör-düğü* (See-NMLZ-POSS) induce a meaning close to *Who he saw*, and even though the meaning is plural, the agreement with such correlative surfaces as singular in Turkish as we discussed in the Introduction. However, it is possible that notional plurality might have resulted in *reverse* attraction effects. We return to the possible explanations for such an effect as well as theoretical implications of the converging null result across both experiments in the General Discussion.

4 General Discussion

The goal of the present study was to investigate how speakers resolve agreement dependencies and what kinds of information they rely on in doing so, using erroneous decisions in a speeded acceptability judgment task as a diagnostic. A better understanding of the mechanisms that give rise to these errors can inform theories of memory retrieval during sentence processing and, more broadly, the architecture of the human sentence processing system. Because the study also compared simplex and complex verbal forms, it bears on theories of morphological processing and the relationship between morphology and syntax.

More specifically, we asked whether surface-form overlap alone can trigger agreement attraction during comprehension. Turkish provides a strong test case because the plural morpheme *-lar* appears on both nouns and nominalized verbs. Moreover, the nominalized verbs in Turkish can serve as a subject by themselves as a headless reduced relative clause. Although both nominal and verbal attractors can occur in subject position, only genitive marked nominal attractors can serve as agreement controllers in Turkish. If surface-form overlap alone is sufficient to trigger attraction, then we should observe attraction from both nominal and verbal attractors. We would also expect comparable attraction effects in both conditions if attraction effects are determined by being similar to a subject. If, however, attraction is gated by abstract controller features rather than surface form, then we should observe attraction only from nominal attractors, which can be controllers, and not from verbal attractors, which cannot. The results strongly supported the latter prediction: robust attraction was observed for nominal attractors but not for verbal attractors, despite maximal surface similarity.

Across two speeded acceptability judgment experiments, nominalized verbal attractors did not induce agree-

ment attraction. In Experiment 1, we observed robust attraction for nominal attractors, closely replicating previous nominal findings (Lago et al., 2019; Türk and Logačev, 2024), but found no attraction for verbal attractors. Crucially, this was not merely a smaller-than-nominal effect: follow-up analyses restricted to the verbal conditions, together with the Bayes factor analysis, provided strong evidence for the absence of verbal attraction. In Experiment 2, which removed the nominal conditions and retained only the verbal ones, the null pattern replicated. Ungrammatical sentences with plural nominalized verbal attractors were accepted no more often than their singular-attractor controls.

This systematic absence of attraction from attractors that share the surface phonology of plural nouns suggests that previously observed effects of phonological overlap, such as reduced reading difficulty and increased accuracy on comprehension questions, do not extend to dependency resolution, contrary to the proposal advanced by Slioussar (2018). The pattern converges with findings from English and Czech showing little to no attraction from pseudo-plural or phonologically plural forms (Bock and Eberhard, 1993; Lacina and Chromý, 2022), but it appears to diverge from findings in Russian (Slioussar, 2018). We propose that the critical factor is not surface overlap per se, but whether the attractor form is associated with being a possible agreement controller (Lago et al., 2019; Bhatia and Dillon, 2022; Bleotu and Dillon, 2024). Russian genitive nouns, despite their surface form, can control agreement in other constructions (Partee and Borschev, 2004), whereas our Turkish verbal attractors and Czech genitive nouns cannot. On this view, controller compatibility, rather than phonological overlap alone, determines whether attraction arises.

In the remainder of the General Discussion, we consider how our results, together with previous findings on case syncretism and phonological overlap, bear on different theoretical accounts of agreement attraction. We then address limitations of the present study and outline directions for future research.

4.1 Accounts of agreement attraction and the role of surface-form overlap

4.1.1 Marking and Morphing Account

Agreement attraction was initially explained through feature percolation: the plural feature of the attractor copies upward—along syntactic projections—to the head of the subject phrase, much like feature inheritance

in formal syntactic theory (Bock and Miller, 1991; Bock and Cutting, 1992; Bock and Eberhard, 1993).

In a phrase such as *the key to the cabinets*, the plural feature of *cabinets* migrates to *key*, erroneously licensing plural agreement on the verb. This early account was refined into the Marking and Morphing model (Eberhard, 1999; Eberhard et al., 2005), which relaxed the requirement that the attractor be in a syntactic percolation path and instead proposed that agreement is computed over a reconciled number value of the subject phrase. Under this model, attraction proceeds in two stages. At the *marking* stage, participants form a conceptual representation of the subject phrase; notional plurality—for instance, the distributive reading available with *The gang on the motorcycles*—can mark the phrase as number-relevant even without a syntactic percolation route. At the *morphing* stage, the number representation of the phrase is reconciled with other sources of number information, weighted by syntactic distance: the further the attractor is from the subject head, the weaker its influence. If the reconciled value is not definitively singular, the verb may be produced or accepted with plural agreement. The model captures fundamental patterns of attraction, including the well-known singular/plural asymmetry—plural attractors, being marked, can spread and shift the reconciled value, whereas singular attractors are underspecified and exert minimal influence (Eberhard et al., 2005). The account was initially developed with production studies in mind. However, Hammerly et al. (2019) subsequently applied the logic to comprehension, arguing that many attraction effects previously attributed to retrieval can be modeled as a product of representational distortion paired with a priori response biases.

What does this account predict for the cases of syncretism and surface-form overlap that are the focus of the current study? Because the model operates over number features—not case features or phonological form—it does not contain a mechanism through which case syncretism, as such, can modulate attraction. The model can predict that a plural attractor should induce some attraction regardless of its case marking, but it does not predict the *additional* boost observed for syncretic attractors in German or Czech (Hartsuiker et al., 2003; Lacina et al., 2025). The spreading mechanism is agnostic to whether the attractor looks like a nominative or not; what it cares about is whether the attractor carries a marked number feature. Importantly, Eberhard et al. (2005) distinguish among three sources of number information: grammatical marking (whether a noun requires plural agreement on verb), morphological number (the presence of plural-like morphology on

inherently singular forms such as *scissors*), and notional number (the conceptual plurality of collective nouns such as *gang* or *fleet*). For the head noun, all three sources contribute to the reconciled number value. For the attractor, however, only grammatical marking—the overt plural morpheme—is relevant; morphological and notional number of the attractor do not participate in the morphing process.

For our verbal attractor manipulation, the prediction depends entirely on how the parser encodes *-lar* on the nominalized verb. If the parser treats this suffix as verbal agreement rather than a number information to be used later, then no inherent plural feature is available to mark the phrase. The morphing pathway is blocked at its source: there is nothing to spread. The model therefore predicts a null result for verbal attractors, consistent with our findings. However, the existing cross-linguistic evidence on syncretism poses a broader challenge for this account. Effects of case syncretism have been repeatedly documented—in German, Czech, and Russian—yet the Marking and Morphing model has no mechanism through which case properties could modulate the spread of number features. It is difficult to reconcile these syncretism-driven effects with the model’s exclusive reliance on grammatical number marking, and to our knowledge, no published extension of the account has attempted to do so.

4.1.2 Noisy Channel and Rational Inference Accounts

Under noisy-channel or rational comprehension approaches, comprehenders do not simply decode the literal input; they infer what the speaker most likely intended, given both the perceived signal and prior expectations about likely sentences (Ryskin et al., 2021; Futrell et al., 2020; Gibson et al., 2013; Levy et al., 2009). If the parser assigns a non-trivial probability to the hypothesis that the input was corrupted—that a singular head was intended as plural, or that words were transposed—then the comprehender may reweight toward an alternative parse that better fits world knowledge or production statistics. Agreement attraction, from this perspective, is a rational response: the parser considers the possibility that the intended sentence contained a plural subject, and the presence of a plural noun nearby raises the posterior probability of that alternative. The likelihood of acceptance of an ungrammatical sentence therefore reflects the probability that a well-formed alternative could have generated the observed string.

More concretely, what matters is the posterior probability of the intended sentence given the observed string.

The question then becomes how likely various intended sentences are and how the parser selects among them. Consider *the key to the cabinets are rusty*: one candidate intended sentence is *the keys to the cabinets are rusty*, which requires only substituting a singular noun for its plural counterpart—a single edit. Another candidate is *the cabinets to the key are rusty*, which requires not only an insertion of plural morphology but also a transposition of the subject and the attractor. The rational inference account predicts that the parser will consider both alternatives, but the substitution route will be favored because it is more similar to the observed string and requires fewer edits.

While [Ryskin et al. \(2021\)](#) aims to account for canonical attraction effects primarily via the substitution route, not all routes to an intended structure carry equal weight. In cases involving phonological overlap or case syncretism, transposition of elements becomes a relevant alternative. Under a transposition analysis, the noisy-channel framework predicts that case-syncretic forms should increase attraction relative to distinctively marked forms, because an attractor whose surface form is compatible with a nominative (controller) parse makes the transposed alternative—in which the attractor and the head swap positions—more plausible. This is precisely the pattern observed in German and Czech. A parallel transposition analysis has been proposed for NPI (negative polarity item) illusions ([Zhang, 2024](#)). For example, in ungrammatical sentences like *Many authors that few critics recommended have ever received...*, [Zhang \(2024\)](#) argues that readers swap the positions of *few* and *many*, inferring that the intended sentence was *Few authors that many critics recommended have ever received...*, which is well-formed.

It is not clear, however, how inference-based accounts would differentiate between the effects of case syncretism and those of phonological overlap. The model, in its current form, should predict similar facilitation from both types of surface similarity, since both increase the plausibility of an alternative intended sentence. Yet the existing cross-linguistic evidence suggests a dissociation. In Czech, [Lacina and Chromý \(2022\)](#) found that only case-syncretic attractors with clear plural marking produced attraction, while phonologically overlapping genitive singulars—whose form happens to match the nominative plural—did not. Similarly, in Russian, case syncretism between accusative and nominative plural forms produced robust attraction, whereas the accidental phonological overlap between genitive singular and nominative plural yielded considerably weaker effects ([Slioussar, 2018](#)). These patterns suggest that the parser is not simply reweighting

toward any alternative sentence consistent with the surface form, but is sensitive to whether the attractor’s morphosyntactic features are genuinely compatible with controllerhood.

4.1.3 Cue-Based Retrieval Accounts

Cue-based retrieval models provide the most explicit mechanistic account of agreement attraction in comprehension (Lewis and Vasishth, 2005; Smith and Vasishth, 2020; Engelmann et al., 2019). In these models, the parser stores each noun phrase as a chunk in content-addressable memory, encoding features such as number, case, gender, and syntactic role. When the verb is encountered, retrieval cues—drawn from the verb’s morphological specification—trigger a parallel search through memory for a chunk matching the profile of an agreement controller. In the case of English subject-verb agreement, for instance, the verb provides cues like [+NOM, +PL], and any chunk partially matching these cues can interfere with retrieval. Attraction arises because a plural attractor partially matches the retrieval cues, and the race dynamics of the retrieval process mean that on some proportion of trials, the attractor is retrieved instead of the true controller (Lewis and Vasishth, 2005; Wagers et al., 2009).

For case syncretism, the canonical model makes a straightforward prediction. Lewis and Vasishth (2005) explicitly proposed that the retrieval cue for case is not abstract case but rather the morphological form of the noun phrase. If the attractor’s surface case is syncretic with the nominative—as in Russian and German, where accusative and nominative forms of certain genders are identical—then the attractor matches the [+NOM] retrieval cue in addition to [+PL], producing stronger interference than a distinctively marked attractor. This aligns with findings in German (Hartsuiker et al., 2003), Czech (Lacina et al., 2025), and Russian (Slioussar, 2018), where syncretic attractors consistently produce more attraction than non-syncretic ones.

How phonological overlap is treated within cue-based retrieval, however, is disputed. The canonical models use morphosyntactic and semantic features to guide memory access (Lewis and Vasishth, 2005; Van Dyke and Lewis, 2003); phonological form influences retrieval only insofar as it determines which morphosyntactic features are assigned to the chunk. Under this assumption, the phonological overlap between verbal *-lar* and nominal *-lar* is irrelevant: if the parser correctly identifies the verbal suffix as agreement marking rather

than nominal plurality, the chunk is encoded without [+PL], and no interference occurs.

Slioussar (2018) proposed a more radical mechanism in which phonological overlap itself can activate competing feature bundles. On her account, a Russian genitive singular noun whose surface form is homophonous with the nominative plural activates not just the chunk corresponding to [+GEN, +SG] but also a competitor chunk carrying [+NOM, +PL]. This co-activated competitor then enters the retrieval race alongside the true target and the veridical encoding of the attractor, increasing the probability of a misretrieval. The mechanism is grounded in well-established principles of spoken word recognition—cohort competition, phonological neighborhood effects—and generalizes naturally from lexical access to syntactic retrieval. If correct, it predicts that any phonological overlap with a controller-relevant form should increase attraction, including the overlap between verbal and nominal *-lAr* in Turkish.

Several problems complicate this prediction. First, Czech exhibits the same genitive singular / nominative plural homophony as Russian, yet Lacina and Chromý (2022) found no attraction from the overlapping forms, and later work by Lacina et al. (2025) confirmed that only case-syncretic—not merely phonologically overlapping—attractors produce attraction. The phonological co-activation mechanism does not explain this cross-linguistic discrepancy without auxiliary assumptions. Second, our Turkish results present the same challenge in a starker form: the overlap between verbal and nominal *-lAr* is not accidental homophony but a genuine identity of form produced by the same phonological string for very similar functional meanings, and yet attraction is entirely absent. Third, prior work on argument structure has shown that phonological similarity between verb forms does not modulate thematic processing (Kush et al., 2015), suggesting that the parser does not routinely exploit phonological overlap during structure building.

A more parsimonious alternative, advanced by Lago et al. (2019) and supported by converging evidence from Hindi and Romanian, is that what the parser tracks is whether the surface form of the attractor is associated with being a possible agreement controller in that language. On this view, the relevant quantity is not phonological similarity but the distributional statistics of the form—how often the morphological shape of the attractor appears as subjects. Lago et al. (2019) argued that the reason Turkish genitive marking induces attraction while English genitive marking does not is that Turkish genitives, but not English ones, regularly

function as subjects in embedded clauses.² Turkish genitives serve as subjects of embedded possessive clauses and control agreement on the embedded verb, establishing a strong statistical link between the genitive form and controllerhood. In our experiments, the verbal attractors in Turkish carry no such association: reduced relative clauses never control matrix verb agreement, and the parser apparently tracks this distributional fact.

This controllerhood account extends naturally to the cross-linguistic pattern. Russian genitive nouns, unlike Czech genitives, can serve as subjects in negative existential constructions and control number and gender marking on modifiers within the noun phrase (Babby, 2001; Partee and Borschev, 2004), providing a distributional basis for their association with controllerhood. Czech genitive nouns lack this association entirely, and no attraction emerges (Lacina and Chromý, 2022). The account also aligns with Romanian findings: Bleotu and Dillon (2024) found that attractors only induced errors when they surfaced with an overt determiner—the form required for an NP to be an agreement controller in Romanian. Bare nouns, which cannot control agreement, did not attract even when they otherwise resembled the subject.

The Hindi data from Bhatia and Dillon (2022) push the account further, however. In Hindi, attraction arose only when the attractor was functioning as an agreement controller within its own clause, regardless of whether it was a subject or an object. Even bare-marked attractors that might be associated with controllerhood due to syncretism did not produce attraction unless they were actively controlling agreement in the sentence. This suggests that Hindi speakers track controllerhood *within the sentence*, not merely language-wide distributional statistics of the form. A similar reinterpretation may apply to German. Hartsuiker et al. (2003) found that accusative-marked plural attractors induced extra attraction beyond the base plural effect, but only for feminine nouns, where the accusative determiner *die* is syncretic with the nominative. In masculine and neuter conditions, the plural accusative attractor also surfaces as *die*, yet no syncretism-driven boost was observed. If German speakers relied on a language-general association between *die* and controllerhood, then the boost should have appeared regardless of the subject head’s gender. The gender-specificity of the

²English genitive pronouns can of course serve as subjects of embedded clauses (*His arrival was delayed*). The critical difference is that English genitive pronouns do not trigger a morphological reflex on the embedded verb—English lacks the possessive agreement paradigm that Turkish has—and so the surface association between genitive marking and controllerhood is far weaker. Lago et al.’s (2019) account can thus be stated more precisely in terms of controller association rather than mere subjecthood following Türk and Logachev (2024).

effect makes more sense if participants are tracking controller-association within the sentence: only when the subject itself surfaces with *die* (as in feminine nominative) can the plural accusative attractor plausibly be taken as a competing controller in that context. When the subject carries *der* or *das*, the mismatch in form blocks this association.

The relationship between keeping track of distributional statistics and dependency resolution was also noticed in English. [Schlueter et al. \(2018\)](#) compared sentences like (20) to their grammatical counterparts. More specifically, they were interested whether conjoined adjectives with a singular head and a singular noun would still induce attraction as much as plural attractors. Across 3 experiments, they found that singular attractors with stacked adjectives (20c) was accepted more often compared to singular attractors without conjoined adjectives (20a). Similarly, the reading times showed that sentences with conjoined adjectives were read faster than sentences with singular attractors and no conjoined adjectives. Both of such effects were smaller than the effects of plural attractors, but they were still reliable. The authors argued that the presence of ‘and’ acted as a weaker cue of plurality. Their results do not directly align with our hypothesized controllerhood-association account, but they do suggest that the parser is sensitive to distributional cues of plurality beyond the presence of an overt plural morpheme.

(20) a. * Singular-Ungrammatical

The slogan about the caring husband were designed to get attention.

b. * Plural-Ungrammatical

The slogan about the caring husbands were designed to get attention.

c. * Conjoined adjectives-Ungrammatical

The slogan about the loyal and caring husband were designed to get attention.

We take no strong position here on whether the parser relies on language-wide distributional statistics or sentence-internal controllerhood tracking. These may represent two ends of a continuum, and different languages may differ in which information is weighted more heavily ([Dillon and Keshev, 2024](#); [Engelmann et al., 2019](#)). What our data do rule out is the claim that surface-form overlap alone—divorced from both controllerhood and syntactic licensing—is sufficient to generate agreement attraction.

4.2 Reservations

Although our results align with the previous cross-linguistic findings and converge on a more parsimonious account of surface-overlap compared to phonological modulation, certain reservations are in order in view of the specific linguistic constructions we are using, the presence of ceiling effects, and the nature of the null effects.

4.2.1 Nominal vs. nominalized verb distinction

Our attractors included genitive modifiers (e.g., *milyonerlerin*, ‘millionaires’) and reduced relative clauses (e.g., *gördükleri*, ‘the one(s) they saw’). If surface-form overlap alone were sufficient to trigger attraction, both types should produce attraction, since they share the same plural morpheme *-lar* and can both surface as subjects. However, the two types of attractors differ in their linguistic status in ways that deserve attention. We believe that although this distinction is an important aspect to control in future experiments, it does not undermine the main conclusion of the paper.

The initial issue concerns the categorial identity of the verbal attractors. Our verbal attractors are nominalized forms: they take case marking, can be modified by adjectival elements, and occupy argument positions—all properties parallel to noun phrases (Göksel and Kerslake, 2005). However, they are also propositional in the sense that their syntactic and semantic properties are derived from their verbal source (Satik, 2024; Demirok, 2019; Lees, 1965; Kornfilt, 1984, 2003; Kornfilt and Whitman, 2011). Demirok (2019) argues that nominalized *-DIK* phrases denote propositions, whereas *-MA* phrases denote events. As illustrated in (21), *-DIK* phrases cannot be subjects of matrix clauses when the predicate denotes an event (21a), but they can when the predicate expresses a judgment about a proposition (22). *-MA* phrases, by contrast, can serve as subjects regardless of predicate meaning (21b).

- (21) a. * [Suzan-ın hata-yı bul-duğ-u] harika bir olay-dı.
Susan-GEN mistake-ACC find-DIK-3SG.POSS amazing a event-PST
Intended: ‘That Susan found the mistake was an amazing event.’
- b. [Suzan-ın hata-yı bul-ma-sı] harika bir olay-dı.
Susan-GEN mistake-ACC find-MA-3SG.POSS amazing a event-PST
‘Susan’s finding the mistake was an amazing event.’

793 (22) [Suzan-in hata-yı bul-duğ-u] {ortada | doğru | biliniyor}.
 Susan-GEN mistake-ACC find-DIK-3SG.POSS {obvious | true | known}
 794 ‘That Susan found the mistake is {obvious | true | known}.’

795 However, concluding that *-MA* is simply more nominal than *-DIK* overlooks an important distinction.
 796 The examples above are cases where *-DIK* encodes a proposition, but the *-DIK* nominalizers used in our
 797 experiment—reduced relative clauses that modify a head noun—are proposed to denote entities rather than
 798 propositions (Demirok, 2019). The strongest evidence for this distinction comes from coordination with *ile*
 799 (‘and/with’), which in Turkish can only coordinate two noun phrases. As shown in (23), propositional *-DIK*
 800 phrases cannot be coordinated with *ile* (23a), but when *-DIK* modifies a noun—that is, in relative clause
 801 formation parallel to our verbal attractors—the resulting phrases can be coordinated (23b), suggesting that
 802 in this use they pattern syntactically as noun phrases.

803 (23) a. * [Suzan’in hata-yı bul-duğ-u] ile [Merve’nin email-i
 Susan-GEN mistake-ACC find-DIK-3SG.POSS with Merve-GEN email-ACC
 804 yaz-dığ-ın-ı] biliyorum.
 write-DIK-3SG.POSS-ACC I.know
 805 Intended: ‘I know that Susan found the mistake and Merve wrote the email.’
 806 b. [Sen-in pişir-diğ-in] ile [ben-im al-dığ-ım] masa-da.
 you-GEN cook-DIK-2SG.POSS with I-GEN buy-DIK-1SG.POSS table-LOC
 807 ‘The thing you cooked and the thing I bought are on the table.’

808 The upshot is that our nominalized verbal attractors does not indicate a necessary confound between differ-
 809 ent categories and cannot be reduced to a simple verb-vs.-noun distinction. Both our nominal and verbal
 810 attractors occupy nominal positions and bear nominal syntax. The *-DIK* forms used in our experiment
 811 are entity-denoting relative clauses—the most noun-like end of the nominalization spectrum. If the parser’s
 812 sensitivity to category were the sole driver of our null result, it would have to operate at a finer grain
 813 than the noun/verb boundary, distinguishing among subtypes of nominal expressions based on their deriva-
 814 tional history. While this remains a logical and empirical possibility, the more parsimonious account is that
 815 controllerhood—not category per se—gates attraction.

816 Even granting this theoretical distinction, the empirical evidence does not clearly support the conclusion
 817 that derived nouns and simple nouns must behave differently in the domain of processing. One might

argue that the parser carries over the verb/noun distinction into derived structures, and that our null effect therefore reflects category sensitivity rather than controllerhood. Existing evidence is consistent with this view: Momma et al. (2020) show that deverbal nouns (*running* as a noun) do not compete with simple verbs (*walking* as a verb) in a sentence-picture interference paradigm, regardless of semantic similarity, suggesting that the parser is sensitive to final categorization (see also Dell et al., 2008). However, work on thematic relations in Italian and Greek points in the opposite direction. Tsaprouni and Manouilidou (2025) show that aspectual mismatches disrupt processing only when the derived verb retains an eventive reading; when the derived word does not carry over eventive properties of the verb, the parser appears to ignore the aspectual mismatch and thus the derivational history of the word (see also Stockall and Manouilidou, 2014). Moreover, Gaston et al. (2020) provide direct evidence that syntactic category does not inhibit activation of category-mismatching lexical items. Using a visual world paradigm in which participants heard sentences with noun-constraining (e.g., “the battleship”) or verb-constraining contexts (e.g., “to bask”) while viewing displays of object pictures only, they found that fixations to noun-only cohort competitors (e.g., *balcony*) increased in both noun and verb contexts. If syntactic category operated via an inhibitory constraint, wrong-category items should have been suppressed in the verb context; instead, they were activated to a comparable degree, indicating that the category constraint operates via facilitation of correct-category candidates rather than inhibition of wrong-category ones. In short, the processing literature does not converge on a single answer: sometimes derivational history matters, sometimes it does not, and the conditions under which derived forms are treated identically to their non-derived counterparts remain poorly understood.

Moreover, regardless of where one lands on the category question, the surface-form properties of our verbal attractors should have provided maximal conditions for phonological co-activation. The *-lAr* suffix on reduced relative clause verbs is string-ambiguous for number: although syntactically it marks agreement with an implicit external plural argument, the surface string does not disambiguate the two functions. A parser encountering *gördükleri* (‘the one(s) they saw’) cannot determine from the string alone whether *-lAr* encodes nominal plural on the entity *taht* was seen or the verbal agreement that the subject of the relative clause is plural, or both. This ambiguity is precisely the kind of surface condition that a phonological co-activation account should exploit—the form *-lAr* is the same morpheme, in the same string position, on a categorially

nominal expression. The broader phonological interference literature supports the expectation that such overlap should produce confusion, whether through transient activation of competing representations in continuous speech processing (McClelland and Elman, 1986; Norris, 1994) or through phonological similarity interference in working memory (Baddeley, 1966; Conrad, 1964; Acheson and MacDonald, 2009). Despite all of this, attraction is absent, setting a strong upper bound on what surface overlap alone can contribute.

4.2.2 Decreased grammaticality with singular verbal attractors

One unexpected finding in Experiment 2 was that grammatical sentences with singular verbal attractors (25) were accepted less often than their plural-attractor counterparts (24). Under any account of attraction, we would expect the opposite pattern: plural attractors should, if anything, reduce acceptability in grammatical conditions by inducing ungrammaticality illusions. We consider two possible explanations for the observed pattern: (i) the singular attractor conditions may have been slightly less natural due to referential properties of singular versus plural reduced relative clauses, and (ii) a possible correlative parse ambiguity in the singular attractor conditions.

(24) Tut-tuk-lar-ı aşçı mutfak-ta sürekli zıpla-dı.
hire-NMLZ-PL-POSS cook kitchen-LOC non.stop jump-PST

‘The cook that they hired jumped in the kitchen non-stop.’

(25) Tut-tuğ-u aşçı mutfak-ta sürekli zıpla-dı.
hire-NMLZ-POSS cook kitchen-LOC non.stop jump-PST

‘The cook that (s/he) hired jumped in the kitchen non-stop.’

The first possibility is that the singular attractor conditions were slightly less natural overall. Singular reduced relative clauses like *tuttuğu* (‘the one [someone] hired’) presuppose a unique referent, whereas plural reduced relative clauses like *tuttukları* (‘the one(s) that they hired’) can be interpreted with an arbitrary subject, often characterized as PRO_{arb} , and are thus referentially less restrictive (Chomsky, 1993; Göksel, 2005). Crucially, plural agreement in Turkish can yield a generic or impersonal reading, effectively downgrading the specificity of the subject referent (e.g., *Burada çok çikolata yerler*, ‘They eat a lot of chocolate here’, where the third-person plural verb conveys an arbitrary, non-referential agent). This property is not unique to Turkish: in Italian, for instance, subjectless third-person plural forms can similarly express a generic

870 interpretation (Cinque, 1988; Mantovan, 2022), as in (26).³

- 871 (26) Qui, *pro* lavor-ano anche di sabato.
here *pro* work-3PL even of Saturday
872 ‘Here they work even on Saturday.’ (adapted from Cinque 1988: 545)

873 This difference may have been amplified by the fact that nearly all of our subject head nouns referred to
874 occupations: participants may have interpreted the plural *-lAr* marking as referring to a non-referential
875 group of people associated with the profession, making the plural attractor conditions pragmatically more
876 felicitous.

877 A second possibility is that *-DIK*-marked reduced relative clauses can receive a correlative parse, as illustrated
878 in (27), taken from Demirok (2017). If participants initially parsed the *-DIK* phrase as a correlative and
879 assigned a ‘who(ever) hired’ reading, this could have introduced some notional plurality into the singular
880 attractor conditions, producing a small but systematic increase in ‘unacceptable’ responses. However, this
881 explanation is unlikely, because plural-marked reduced relative clauses can equally receive a correlative
882 parse: if correlative ambiguity were responsible for the observed pattern, it should have appeared in the
883 plural attractor conditions as well, which it did not.

- 884 (27) Mary [John-un pişir-diğ-in-i] ye-di.
Mary John-GEN cook-NMLZ-3SG-ACC eat-PST
885 ‘Mary ate what John cooked.’

886 In either case, the key point is that the observed pattern does not undermine our main conclusion about the
887 absence of attraction from verbal *-lAr*. The singular attractor conditions were still accepted at a very high rate
888 (over 90%), and the critical comparison between singular and plural attractors within the ungrammatical
889 conditions shows no evidence of attraction. The unexpected pattern in the grammatical conditions may
890 reflect subtle pragmatic or parsing factors, but it does not affect the interpretation of the null effect for
891 verbal attraction.

³We thank Furkan Dikmen for drawing our attention to this parallel.

4.2.3 English pseudoplurals and how they fit

An important starting point for our study was the observation that English pseudoplurals—singular nouns whose phonological form coincidentally resembles a plural—do not induce agreement attraction. Bock and Eberhard (1993) showed that with preambles like *The player on the courts ...*, the plural noun *courts* induces attraction, but with preambles like *The player on the course ...*, the singular noun *course* does not, despite sharing the final /s/ with the plural form. Haskell and MacDonald (2003) replicated this pattern, and Nicol et al. (2016) extended it to comprehension.

These findings align with the controllerhood account we have been developing. Consider the paradigm used by Nicol et al. (2016), illustrated in (28). Plurality on the object noun *gardens* induces attraction when (28b) is compared with (28a), and when (28d) is compared with (28c). However, plurality on the possessor *elves*’ does not induce additional attraction: the comparison of (28c) with (28a), and of (28d) with (28b), yields no reliable increase. As Lago et al. (2019) argued, this pattern follows if the parser tracks distributional statistics of controllerhood: English possessive constructions do not establish a strong association between genitive marking and controllerhood, whereas Turkish genitives—which regularly control agreement on embedded verbs—do.

- (28) a. The statue in the elf’s garden ...
b. The statue in the elf’s gardens ...
c. The statue in the elves’ garden ...
d. The statue in the elves’ gardens ...

A separate question concerns why English pseudoplurals themselves fail to attract, given that their surface form is phonologically identical to a plural noun. The word *course* /kɔːɪs/ ends in the same sibilant as the plural *courts* /kɔːɪts/, yet it does not trigger attraction. One possible explanation is that *course* cannot be decomposed into a stem plus a plural morpheme: there is no parse of /kɔːɪs/ as /kɔːɪ/ + /s/ that yields a recognizable stem, unlike Russian genitive singulars where the surface form is segmentable into stem plus a case ending that is syncretic with the nominative plural. Bock and Eberhard (1993) offered a different account, suggesting that the avoidance of producing two adjacent sibilants—/s/ in *course* followed by /z/

in *is*—might itself contribute to some attraction effects in English, such that speakers produce *the key to the cabinets are* partly to avoid the sibilant cluster in *the key to the cabinets is*. However, this phonotactic account does not generalize to the cross-linguistic patterns we have been discussing, and the broader evidence favors a controllerhood-based explanation.

We leave open the question of whether English pseudoplurals that can be accidentally segmented into a recognizable stem and a plural-like suffix might induce attraction. For instance, the word *ax* / $\sim$ ks/ is phonologically very similar to *acts* / $\sim$ kts/ in certain dialects; our account would predict that a singular noun like *ax* might be more likely to induce attraction in contexts where an *act* + /s/ parse is available. This remains an interesting avenue for future research.

4.3 Interpreting Null Effects as a Primary Result

The core theoretical claim of this paper rests on a null effect—the absence of attraction from verbal attractors—and so the evidential standards applied to this null matter. We treated the absence of verbal attraction as a target estimand rather than a by-product of non-significance, and three lines of evidence support this treatment.

First, the null replicates across experiments with different condition structures. In Experiment 1, verbal attractors were interleaved with nominal attractors that produced robust attraction; in Experiment 2, only verbal conditions were retained. The verbal null was the same in both designs. This cross-design replication is important because it rules out the possibility that the null arises from a specific feature of one experimental context—for instance, that exposure to strong nominal attraction in Experiment 1 might have masked a weak verbal effect. Experiment 2 removes nominal items entirely, and the null persists.

Second, the null is coherent across descriptive and model-based analyses. In both experiments, ungrammatical verbal conditions show very similar acceptance rates across attractor number, and the model-implied estimates for verbal attraction are centered near zero. Furthermore, the Experiment 1 follow-up analysis restricted to verbal trials directly addresses the most relevant question: whether verbal attraction is present at all, not merely whether it is smaller than nominal attraction. That analysis, too, found no evidence for

attraction.

Third, the studies are well-powered and the null is quantified with formal model comparison. Experiment 1 included 92 participants and Experiment 2 included 78 participants—sample sizes that are substantially larger than those in prior Turkish attraction studies (Lago et al., 2019; Türk and Logačev, 2024) and comparable to or exceeding those in most published attraction experiments cross-linguistically. To move beyond a simple absence of significance, we computed Bayes Factors via bridge sampling for the verbal-only follow-up model in Experiment 1, comparing a model that included the critical Grammaticality $\times$ Attractor Number interaction to one that omitted it. The resulting $BF_{01} = 6.13$ indicated moderate evidence in favor of the null model—that is, the data were approximately 6 times more likely under a model without verbal attraction than under one with it. Crucially, because Bayes Factors are sensitive to the prior distribution, we conducted a prior-sensitivity analysis in which we refit the same model pair under five increasingly concentrated zero-centered normal priors on the fixed effects ($\mathcal{N}(0, 2.0)$ through $\mathcal{N}(0, 0.1)$). The null was favored across all five priors (BF_{01} ranging from 2.73-12.40), confirming that the conclusion does not depend on a particular prior specification.

Fourth, the null does not imply that the true effect is mathematically zero. It implies that any residual effect, if present, is not strong enough to support the theoretical prediction that surface-form overlap alone should generate attraction. The burden of evidence for a phonological co-activation account is now on demonstrating conditions under which verbal overlap *does* produce attraction—conditions that the present design was specifically constructed to provide and that failed to yield an effect.

Two additional methodological concerns deserve comment. One is whether speeded acceptability judgments might miss transient processing interference that dissipates before the final decision. This is possible in principle and motivates future work with time-sensitive measures such as self-paced reading, eye-tracking, or EEG. However, the present designs were optimized for detecting attraction of the size observed in prior Turkish studies (Lago et al., 2019; Türk and Logačev, 2024), and the null still constrains theory: any transient effect would have to be weak enough not to influence final acceptability under these conditions. A second concern is whether participants might simply have ignored verbal morphology altogether—a shallow-processing ac-

count. This is unlikely for two reasons: participants showed robust sensitivity to verbal agreement in the baseline grammaticality contrasts, reliably rejecting ungrammatical sentences, and Experiment 2 revealed sensitivity to attractor number *within* grammatical conditions, demonstrating that morphological variation on the reduced relative clause was being registered. The null result for verbal attraction therefore cannot be attributed to a general failure to process the attractor’s morphology.

4.4 Response Bias

One concern with speeded acceptability judgment tasks is that participants differ in their overall willingness to press ‘acceptable’—a response bias that is orthogonal to perceptual sensitivity. Hammerly et al. (2019) and Türk (2022) have shown that filler composition and task instructions can shift overall acceptance rates, and that such shifts can obscure the true effect of experimental manipulations in speeded judgment paradigms. Our Experiment 1 data are consistent with such a shift: overall acceptance rates for nominal-attractor sentences were lower than in Türk and Logačev (2024), suggesting that the inclusion of non-attracting verbal trials shifted the response criterion toward a more conservative baseline. Importantly, however, this shift did not substantially reduce the *magnitude* of attraction within the nominal conditions. Exposure to non-attracting verbal trials shifted overall bias without recalibrating the attraction effect itself.

A remaining question is whether individual differences in response bias might explain the absence of verbal attraction—that is, whether participants who adopted a more liberal criterion showed verbal attraction effects that were washed out in the group-level analysis. Although we did not design the experiments to test this directly, we conducted a post-hoc analysis relating each participant’s response criterion c to their attraction magnitude, defined as the difference in false alarm rates between plural- and singular-attractor ungrammatical conditions. For each participant, we computed the hit rate (accepting grammatical sentences) and false alarm rate (accepting ungrammatical sentences) separately for each attractor type, and from these derived the signal detection criterion $c = -0.5 \times [z(\text{hit rate}) + z(\text{false alarm rate})]$.

Figure 5 shows subject-level attraction estimates plotted against response criterion for each condition. For nominal attractors in Experiment 1, response criterion was reliably correlated with attraction magnitude ($r = -0.43, p < .001$): participants with a more liberal criterion (lower c) tended to show larger attraction effects,

consistent with the idea that response bias modulates the observable magnitude of attraction when attraction is genuinely present. For verbal attractors, by contrast, no such relationship emerged. In Experiment 1, the correlation between criterion and verbal attraction was negligible ($r = -0.12$, $p = 0.242$), and the same held in Experiment 2 ($r = -0.10$, $p = 0.391$). The absence of a criterion–attraction relationship in the verbal conditions across both experiments indicates that the null effect for verbal attraction is not an artifact of conservative responding: even participants who adopted a liberal criterion and accepted ungrammatical sentences at high rates did not show verbal attraction.

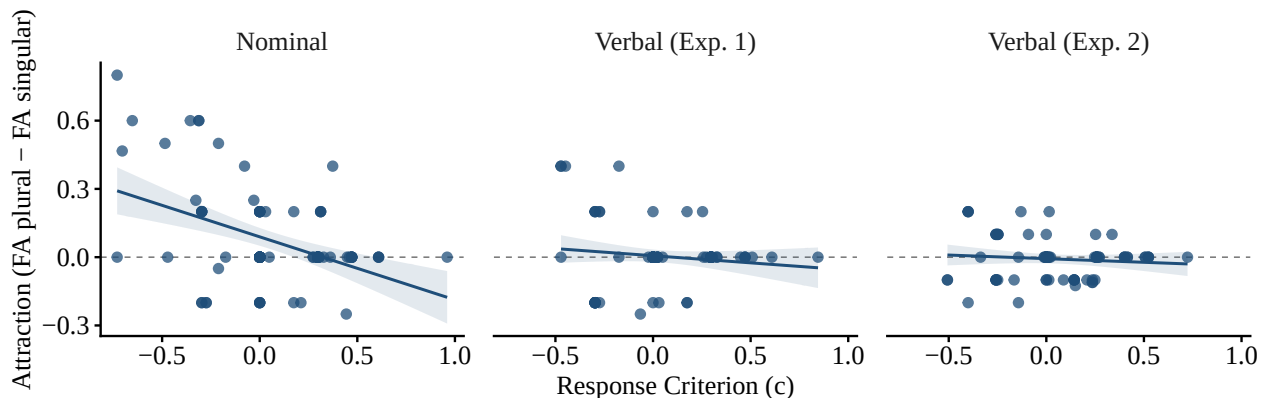


Figure 5: Subject-level attraction estimates as a function of response criterion, shown separately for nominal attractors and verbal attractors in Experiment 1 and Experiment 2. Points represent participants; lines show linear fits with confidence bands.

5 Conclusion

The central finding is a dissociation under minimal surface contrast: nominal attractors produce attraction, verbal attractors with the same plural surface form do not. This null result is theoretically informative because the design maximized the conditions under which phonological co-activation should have produced an effect—the attractor is categorially nominal, the *-lar* suffix is string-ambiguous for number, and phonological overlap in memory is well-established as a source of interference. The absence of attraction under these conditions indicates that surface-form overlap alone is insufficient to drive agreement illusions.

Several theoretical frameworks converge on this conclusion. Under cue-based retrieval, the verbal attractor either fails to match the critical retrieval cues (because agreement features are encoded differently from inherent number) or fails to carry the distributional association with controllerhood that enables a form

to compete during retrieval. Under Marking and Morphing, there is no spreadable number feature. Under noisy-channel approaches, there is no well-formed alternative parse to reweight toward. In each case, the null result follows from the asymmetry between forms that can and cannot function as agreement controllers.

Agreement attraction in comprehension is therefore better characterized as grammar-constrained memory retrieval than as form-driven confusion. The human parser is sensitive to fine-grained distinctions in the morphosyntactic status of potential competitors—distinctions that are invisible on the surface of the string.

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